

Volatility Trading Case

Rotman International Trading Competition (RITC)x

The 'Volatility Trading Case', Copyright © Rotman Finance Research and Trading Lab, all rights reserved. This case has been adapted from the 'Options Trading 3 (OP3) Case', developed for the RIT Market Simulator platform (rit.rotman.utoronto.ca)

OVERVIEW

The Volatility Trading Case gives participants the opportunity to generate profits by implementing options strategies to trade volatility. The underlying asset of the options is a non-dividend-paying Exchange Traded Fund (ETF) called RTM that tracks a major stock index. Participants will be able to trade shares of the ETF and 1-month call/put options at 10 different strike prices. Information including the ETF price, options prices, and news releases will be provided. Participants are encouraged to use the provided information to identify mispricing opportunities and construct options trading strategies accordingly.

DESCRIPTION

The Volatility Trading Case will be 5 minutes long and represent one month of trading.

Participants may choose how they execute their trading strategy during the case. They may trade manually through the RIT Client, build and use a model to support their trading decisions, or develop an automated trading algorithm using the RIT API. Participants are also permitted to modify their models or algorithms throughout the case. A base template model will be provided for participants who wish to use or adapt it, although participants are encouraged to develop their own models and algorithms where appropriate.

MARKET DYNAMICS

Participants will be able to trade RTM and 10 separate options contracts on RTM. All options are European, so early exercise is not allowed.

Call Ticker	Strike Price	Put Ticker
RTM48C	48	RTM48P
RTM49C	49	RTM49P
RTM50C	50	RTM50P
RTM51C	51	RTM51P
RTM52C	52	RTM52P

Information including the stock price, option prices, and news releases are provided throughout the case.

The price path of RTM is generated using the following (martingale) process:

$$P_{RTM,t} = P_{RTM,t-1} * (1 + r_t) \text{ where } r_t \sim N(0, \sigma_t)$$

The price of the stock is based on the previous price multiplied by a return that is drawn from a normal distribution with a mean of zero and standard deviation (volatility) of sigma.

RTM is a very popular ETF and its options are priced by a very large market-maker who will always quote a bid/ask spread of 2 cents (e.g. \$50.00/\$50.02 for the RTM, or \$4.11/\$4.13 for the RTM49C). There is enough market depth associated with the bids and asks such that any market order can be immediately executed without price impact (there are no liquidity constraints in this case).

Your analysts have developed a very effective model to forecast the realized volatility of the underlying security RTM; they will be sending you weekly updates. The market maker quotes prices for options using the Black-Scholes pricing model and a variance risk premium equal to zero. (The start-up model also assumes a risk-free interest rate of 0% but your instructor may change this assumption). Therefore, in equilibrium the implied volatility associated with an options price will be equal to the realized volatility of the underlying security.

However, to open trading opportunities based on mispriced options, we assume that the volatility forecasts made by the market maker are less informed than those provided by the informed analysts. Therefore, after a discrete change in the realized volatility of the underlying, the market maker's volatility forecast embodied in the quoted options price will not accurately reflect the volatility of the underlying security (RTM). Nevertheless, as time goes on, the market maker learns about the underlying volatility – at least until a new shock to the realized volatility occurs.

Your job is to look for mispricing opportunities in the options markets and profit from that mispricing. When mispricing occurs, you should take a position in the options and at the same time trade the underlying security to hedge your delta exposure in order to build a delta neutral portfolio. For example, if a call option has an implied volatility that is 5% greater than the analysts' forecasted realized volatility of the ETF, the option is overpriced. In this case, you should write call options and then hedge the delta exposure by trading the underlying asset appropriately.

For ease of computation, one trading year is assumed to be 240 days (20 days $\times$ 12 months). The case represents one month (20 days) of calendar time. The trading period is 5 minutes (300 seconds) and is divided into 4 weeks, with $t=1\dots75$ being week one, $t = 76\dots150$ week two, and so on. At the beginning of each week, the volatility value (sigma) will shift, and the new value will be provided to the students. In addition, at the middle of each week (e.g. $t=37$) an estimate of next week's volatility value will be announced.

The Chief Risk Officer (CRO) at your firm also wants you to focus on trading volatility without exposing your position to changes in the price of the underlying. That is, the CRO requires you to hedge the delta of your portfolio. Recognizing the transactions costs of delta hedging continuously, you are asked to keep the delta of your portfolio between -7000 and 7000. The risk management office (CRO) will monitor your position throughout the month and will charge you a penalty for every second that you exceed the delta limit (± 7000), that is:

$$\text{Penalty for second } t = (|\text{Portfolio's Delta at time } t| - \text{Delta Limit}) \times x$$

in which *Delta Limit* is the limit imposed by the CRO and x is the fine rate set at \$0.10.

Sample News Release Schedule

Time	Week	Release
1	Week 1	The realized volatility of RTM for this week will be 20%
75	Week 1	The realized volatility of RTM for next week will be between 27-30%
150	Week 2	The realized volatility of RTM for this week will be 29%
...	...	...
450	Week 4	The realized volatility of RTM for this week will be 26%

TRADING/POSITION LIMITS AND TRANSACTION COSTS

Each participant will be subject to gross and net trading limits specific to the security type as specified below. The gross trading limit reflects the sum of the absolute values of the long and short positions across all securities; the net trading limit reflects the sum of long and short positions such that short positions negate any long positions. Trading limits will be enforced and participants will not be able to exceed them.

Security Type	Gross Limit	Net Limit
RTM ETF	50,000 Shares	50,000 Shares
RTM Options	2,500 Contracts	1,000 Contracts

The maximum trade size will be 10,000 shares for RTM and 100 contracts for RTM options, restricting the volume of shares and contracts transacted per trade to 10,000 and 100, respectively. Transaction fees will be set at \$0.02 per share traded for RTM and \$2.00 per contract traded for RTM options. As with standard options markets, each contract represents 100 shares (purchasing 1 option contract for \$0.35 will actually cost \$35 plus a \$2 commission, and will settle based on the exercise value of 100 shares).

POSITION CLOSE-OUT

Any outstanding position in RTM will be closed at the end of trading based on the last-traded price. There are no liquidity constraints for the options nor RTM. All options will be cash-settled based on the following:

$$\text{Call Option Payout} = \max\{0, S - K\}$$

$$\text{Put Option Payout} = \max\{0, K - S\}$$

Where,

S is the last price of RTM;

K is the strike price of the option.

KEY OBJECTIVES

- Build a model to forecast the future volatility of the underlying ETF based on known information and given forecast ranges. Participants should use this model with an options pricing model to determine whether the market prices for options are overvalued or undervalued. They should then trade the specific options accordingly.
- Use Greeks to calculate the portfolio exposure and hedge the position to reduce the risk of the portfolio while profiting from volatility differentials across options.
- Seek arbitrage opportunities across different options.